Gaurav Pandey

Curriculum Vitae

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Education

2022 – Present **Doctor of Philosophy - PhD**, *National Taipei University of Technology*, Taipei, Taiwan

2017 – 2019 **Master's, Mechanical Engineering**, *Chang Gung University*, Taiwan

2012 – 2016 **Bachelor's, Mechanical Engineering**, *A.P.J. Abdul Kalam Technical University*, U.P, India

Publications

1. **Pandey, Gaurav**, et al. "X-ray metrology for advanced semiconductor manufacturing of next-generation logic devices." *Journal of Micro/Nanopatterning, Materials, and Metrology* 25.1 (2026): 010901-010901. **(Journal article)**
2. **Pandey, Gaurav**, et al. "Emerging x-ray metrology techniques with evolving front-end-of-line high-technology IC nodes." *Metrology, Inspection, and Process Control XXXIX* 13426 (2025): 311-318. **(International Conference)**
3. Chein, W. H., **Pandey, G.**, Das, S., & Chen, L. C. (2024, June). Advancements in metrology for advanced semiconductor packaging. In *Optics and photonics for advanced dimensional metrology III* (Vol. 12997, pp. 127-142). SPIE **(International Conference)**

Experience

Mar 2023 – Present **Research Intern**, *Center of Measurement & Standards, ITRI*, Hsinchu, Taiwan
- Conduct research on X-ray metrology for front-end-of-line (FEOL) semiconductor characterization.
- Develop measurement and analysis approaches for critical-dimension and structural characterization of advanced logic devices, including gate-all-around nanosheet architectures.
- Contribute to X-ray instrumentation and vacuum-system development for soft and tender X-ray measurements.

Sep 2022 – Present **Research Assistant**, *Precision Metrology Lab, NTU*, Taipei, Taiwan
Conduct research focused on understanding optical critical dimension (OCD) metrology and automated optical inspection (AOI) applications.

Feb 2022 – Jun 2022 **Product Design and Development**, *Botlab Dynamics, TBIU, IIT Delhi*, New Delhi
Directed end-to-end design and development of light-show drone fleets across multiple weight classes, scaling from rapid 3D-printed prototypes to mass production.

Aug 2021 – Feb 2022 **Design & Application Engineer**, *Aerogram, TBIU, IIT Delhi*, New Delhi
Responsible for the product development cycle (PDC) of various IoT monitoring devices with targets to meet the criteria of certifications like ATEX and ICAT.

Nov 2019 – Mar 2021 **Asst. Manager-Design & Development**, *Thousand Perfection Melamine Co. Ltd*, Guangdong Province, Dongguan, China
Worked in the R&D division to design and develop compressive mold tools for new products.Part of the team to facilitate methods to monitor and control the critical points of different forming load machinery.

Skills

Programming Python, Matlab, C, Arduino

CAD/CAE Solidworks, UG. NX, Proteus, ANSYS, QForm

Languages

Hindi, English, Mandarin